

CPIS-351 IS Analysis and Architecture Design

Lab Description

This course is another important course in the series of software engineering courses, which has the pre-requisite CPIS-250. Importantly this course is deal with the design phase activities using structural and object oriented techniques. System architecture, traditional approach to design, object oriented approach to design, are some major topics covers in this lab.

Lab Schedule

Lab/Tutorial 90 minutes per week.

Lab Requirements

MS Project Management
MS Visio
Argo UML

Lab Resources

FCIT-made lab manual
Set of handouts (e.g. Case studies)

Topics Coverage

Weeks	Topic
1	Introduction of Lab and Tools
2	Traditional Approach to Design (Part-1)
3	Traditional Approach to Design (Part-2)
4	Object Oriented Approach to Design (Part-1)
5	Lab Activity – Class Diagram
6	Object Oriented Approach to Design (Part-2)
7	Interaction Communication Diagram
8	Activity and State Machine Diagram
9	UML Component Diagram
10	UML Deployment Diagram
11	UML Package Diagram
12	Discussion and Case Study
13	Project Report Submission and Presentation

Lab Learning Outcomes (LLO)

By completion of the lab the students should be able to

1. Tools expertise
2. Know about basics of this course
3. Difference between Object oriented and traditional approach of system analysis and design
4. Know about use and how to design data flow diagram
5. Know about use and how to design system flowchart
6. Understand the structure chart
7. Design the structure chart
8. Develop sequencing of program using pseudocode
9. Understand the class and domain diagram
10. Convert domain diagram into class diagram
11. Think critically in designing use case diagram
12. Convert scenarios in sequence diagram
13. Linking of the Objects via Links, Messages and Transactions
14. Develop the state machine diagram
15. Develop UML component diagram
16. Develop UML deployment diagram
17. Use three-layer approach for building package diagram

Lab Assessment

Week	Assessment	Grade %
5	Graded Lab Work-1	2.5
6	Graded Lab Work-2	2.5
8	Graded Lab Work-3	2.5
12	Graded Lab Work-4	2.5
Final Week	Group Project Submission	15

Coordinator(s)

Lab Instructor(s)